

Femto Cells Connecting Fixed and Mobile

How does an existing cellular system work? And how will the emerging femto cells enable improved indoor coverage and fixed-mobile convergence? Find out



Picochip PC302 residential femtocell enables better communication indoors

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In cellular systems, one primary objective is to use the spectrum efficiently. In order to use the limited radio spectrum efficiently, the same frequencies are reused in non-adjacent cells.

Cellular radio systems were implemented for the first time in the advanced mobile phone system (AMPS) in the USA. In AMPS, the concept of frequency reuse was introduced to support more users. AMPS is an analogue system known as the first-generation cellular radio system.

Current cellular radio systems are known as the second-generation (2G) systems. These are digital. In the USA, two standards are used for 2G systems:

IS-95 (CDMA) and IS-136 (D-AMPS). Europe and Asia implemented a system called the global system for mobile communications (GSM). Japan uses a system called personal digital cellular (PDC).

GSM is used in more than a hundred countries by over 215 operators inside and outside of Europe. The Japanese PDC system is the second largest digital cellular system, followed by the

IS-54/136 and IS-95 systems used in North America.

The third generation of cellular systems (3G systems) will allow different systems to interoperate in order to attain global roaming across different cellular radio networks with full man and machine mobility and unique single identification number (UTN: universal telecommunication number) globally.

The International Telecommunication Union (ITU) has been doing research on 3G systems since the mid 1980s and a version of a 3G system is called International Mobile Telecommunications - 2000 (IMT-2000).

3G systems have the following major objectives: